

On the Correctness of Real Parameterization and Conditional Inference under the Binary Normalization Hypothesis

Problem Statement

We consider the equalities

$$\begin{aligned}x^n &= (m^n - p^n)^n, \\z^n &= (m^n + p^n)^n, \\y^n &= 2n l^n,\end{aligned}$$

where the parameters

$$m, p, l$$

are not required to be integers, but can be real numbers.

We also consider a further conditional step: if we accept as a separate hypothesis, or an unproven axiom, the equality

$$o(n) = 2,$$

is the subsequent inference

$$2^n = 2n$$

correct, and therefore

$$n = 1 \quad \text{or} \quad n = 2.$$

1. Correctness of the Equalities for x^n and z^n

The equalities

$$x^n = (m^n - p^n)^n$$

and

$$z^n = (m^n + p^n)^n$$

are correct provided that we have previously set

$$x = m^n - p^n,$$

$$z = m^n + p^n.$$

This is the usual raising of equal quantities to the same power.

In other words, if

$$x = m^n - p^n,$$

then automatically

$$x^n = (m^n - p^n)^n.$$

Similarly, if

$$z = m^n + p^n,$$

then automatically

$$z^n = (m^n + p^n)^n.$$

Here it is not required for m and p to be integers. It is sufficient that the corresponding expressions are defined in the numerical domain under consideration, for example, in the domain of real numbers.

2. Correctness of the Equality $y^n = 2n l^n$ for a Real l

The equality

$$y^n = 2n l^n$$

is correct if l is understood as a real scaling parameter.

After the binomial expansion of the expression

$$(m^n + p^n)^n - (m^n - p^n)^n$$

we obtain a sum over only the odd binomial terms:

$$(m^n + p^n)^n - (m^n - p^n)^n = 2 \sum_{j \text{ odd}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

Let us denote the inner sum by

$$S = \sum_{j \text{ odd}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

Then

$$y^n = 2S.$$

If we now define the real parameter l by the condition

$$l^n = \frac{S}{n},$$

then we obtain

$$S = n l^n.$$

Consequently,

$$y^n = 2S = 2n l^n.$$

Thus, the equality

$$y^n = 2n l^n$$

is correct as a real representation when defining l via

$$l^n = \frac{1}{n} \sum_{j \text{ odd}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

At the same time, it is not required for l to be an integer.

3. Important Difference from Integer Divisibility

If l is considered as a real number, it is not necessary to assert that the sum

$$S = \sum_{j \text{ odd}} \binom{n}{j} (m^n)^{n-j} (p^n)^j$$

is divisible by n in integers.

This is a different, stronger statement.

For real parameterization, it is sufficient to define

$$l^n = \frac{S}{n}.$$

Therefore, a potential problem with integer divisibility does not destroy the real scheme, provided it is explicitly stated that

$$l \in \mathbb{R}.$$

4. Transition to the Scaling Parameter o

If we introduce the scaling parameter o via the equality

$$y = o l,$$

then for $l \neq 0$ we have

$$y^n = o^n l^n.$$

On the other hand, from the previous equality

$$y^n = 2n l^n.$$

Consequently,

$$o^n l^n = 2n l^n.$$

If

$$l \neq 0,$$

then we can divide by l^n and obtain

$$o^n = 2n.$$

In other words,

$$o = (2n)^{1/n}$$

given the chosen positive real branch of the root.

5. The Conditional Step $o(n) = 2$

Next, a separate binary normalization hypothesis is introduced:

$$o(n) = 2.$$

It is important to emphasize that this is precisely a hypothesis, not an automatically proven fact.

By accepting this hypothesis, from

$$o(n)^n = 2n$$

we obtain

$$2^n = 2n.$$

This is already a purely elementary exponential-linear equation.
It is equivalent to

$$n = 2^{n-1}.$$

Its positive integer solutions are:

$$n = 1$$

and

$$n = 2.$$

For all

$$n > 2$$

we have the strict inequality

$$2^n > 2n.$$

Therefore, under the condition

$$o(n) = 2$$

the exponents

$$n > 2$$

are excluded.

6. The Meaning of the Verification in Coq

The fifth version of the Coq file formalizes exactly this real scheme.

It introduces a structure where the parameters

$$m, p, l, x, y, z$$

are treated as real numbers.

In this structure, the equalities

$$x^n = (m^n - p^n)^n,$$

$$z^n = (m^n + p^n)^n,$$

$$y^n = 2n l^n$$

are specified as data fields.

Furthermore, it is proved that if additionally

$$y = o l$$

and

$$l > 0,$$

then from

$$y^n = 2n l^n$$

it follows that

$$o^n = 2n.$$

This corresponds exactly to the condition

$$\text{covers_with } o \ n.$$

7. Consistency Between the Fourth and Fifth Versions

The fourth version of the Coq file verifies the main conditional scheme:

if a hypothetical solution implies $o^n = 2n$,

and if it is then assumed that

$$o = 2,$$

then the only possibilities are

$$n = 1 \quad \text{or} \quad n = 2.$$

The fifth version extends the fourth.

It adds an explicit real structure for the parameters

$$m, p, l, x, y, z$$

and formally shows the transition

$$y = ol, \quad y^n = 2n l^n \quad \implies \quad o^n = 2n.$$

Therefore, the fourth and fifth versions are consistent with each other.

The fifth version does not contradict the fourth, but clarifies and extends it by adding a real level of parameterization.

Final Conclusion. *Under the understanding that*

$$m, p, l \in \mathbb{R}$$

the equalities

$$x^n = (m^n - p^n)^n,$$

$$z^n = (m^n + p^n)^n,$$

$$y^n = 2n l^n$$

are correct as a real formal scheme.

The fifth version of the Coq file consistently formalizes this scheme using real parameters.

If we then separately accept the binary normalization hypothesis

$$o(n) = 2,$$

the further inference

$$2^n = 2n$$

and the restriction

$$n \in \{1, 2\}$$

are correct.

Consequently, under the specified formulation, the underlying conditional logic is consistent: the weak point lies neither in the real parameterization nor in the equality

$$y^n = 2n l^n,$$

but exclusively in the status of the hypothesis

$$o(n) = 2.$$